

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourg Public Administration — Civil Servant NLU Deployment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Severe structural misalignment between benchmark output ontology and deployment requirements. All eight LTZGLUE tasks produce single hard labels [Q67] across narrow label spaces: binary headline acceptability [Q148], three-class sentiment [Q149], binary/four-class linguistic acceptability [Q33, Q151], BIO-tagged NER over PER/ORG/LOC/MISC/DATE [Q44, Q152], five-class topic [Q48, Q153], nine voice-assistant intents [Q154], and binary entailment [Q155]. The deployment explicitly requires multi-label classification with confidence scores for routing and human-fallback escalation [elicitation Q4], and demands administrative topic categories (cross-border worker, housing, transport, tax, communal vs. national) and urgency/frustration scales — none of which appear in any task. Sentiment's three-class scheme [Q149] cannot encode urgency or formal-but-understated frustration cues that are operationally significant per the elicitation [elicitation Q3] and regional cultural notes. EU AI Act considerations [WEB-12] and the language law's requirement to detect citizen language choice [WEB-14] introduce further required output dimensions (language-of-input, possibly risk flags) absent from the benchmark. OO is HIGH priority. This is the most severe dimension violation.

ID	Evidence
Q67	"Together, the eight tasks in LTZGLUE form a broad and balanced evaluation suite, covering four binary and four multi-class settings, sentence- and document-level inputs, as well as a token-level sequence-labelling task." (p.5)
Q148	"headline_classification: Decide if the given title/headline fits the text. Output True or False." (p.15)
Q149	"sentiment_analysis: Classify sentiment of the text. Allowed labels: positive, neutral, negative." (p.15)
Q33	"The binary dataset distinguishes between correct (1) and incorrect (0), for which the label 0 encompasses the categories Verb, Adj, Syntax and Ortho." (p.4)
Q151	"linguistic_acceptability_multilabel: Detect if the sentence is correct or if some element is wrong. If the sentence is correct, Output: correct. If it is not, Output the label referencing the wrong element: syntax, verb, ortho or adj." (p.15)
Q44	"To ensure compatibility, the tag set is harmonised by merging the GPE and LOC categories into a single location label, while retaining PER, ORG, and MISC unchanged." (p.4)
Q152	"ner: Perform Named Entity Recognition on the given sequence of sentence tokens. Output tags as lists of ner_tags. Allowed Tags: O, B-LOC, I-LOC, B-PER, I-PER, B-DATE, I-DATE, B-ORG, I-ORG, B-MISC, I-MISC." (p.15)
Q48	"From the available categories, we focused on five principal domains: SPORTS, CULTURE, TECHNOLOGY, BUSINESS, and ANIMALS." (p.4)
Q153	"topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology." (p.15)
Q154	"slot_intent_detection: Detect the intent for the text given. Allowed intents: reminder/show_reminders, weather/find, reminder/set_reminder, reminder/cancel_reminder, alarm/snooze_alarm, alarm/show_alarms, alarm/set_alarm, nalarm/cancel_alarm, nalarm/time_left_on_alarm." (p.15)
Q155	"recognizing_textual_entailment: Determine if the information in the second sentence is entailed in the first one. Output: 0 or 1." (p.15)
Q4	"Small and under-researched languages are particularly difficult to evaluate, as is the case with Luxembourgish (LTZ), the national language of Luxembourg, with around 400k speakers." (p.1)
Q3	"We evaluate various pre-trained language models for LTZ to present an overview of the current capabilities of these models on the LTZ language." (p.1)

WEB-12	EU AI Act Annex III Category 5(a) high-risk classification potentially relevant — implies need for documented decision rules and oversight outputs not in benchmark (https://artificialintelligenceact.eu/annex/3/)
WEB-14	Language law requires response in citizen's chosen language — implies language-of-input output dimension absent from benchmark (https://data.legilux.public.lu/filestore/eli/etat/leg/loi/1984/02/24/n1/jo/fr/html/eli-etat-leg-loi-1984-02-24-n1-jo-fr-html.html)

Strengths

- Standard, well-documented label schemas (BIO for NER, ternary sentiment) that are interoperable with downstream pipelines [Q44, Q149, Q152]
- NER label set (PER, ORG, LOC, MISC, DATE) is partially useful for entity extraction within administrative messages [Q152]

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Q44	"To ensure compatibility, the tag set is harmonised by merging the GPE and LOC categories into a single location label, while retaining PER, ORG, and MISC unchanged." (p.4)
Q149	"sentiment_analysis: Classify sentiment of the text. Allowed labels: positive, neutral, negative." (p.15)
Q152	"ner: Perform Named Entity Recognition on the given sequence of sentence tokens. Output tags as lists of ner_tags. Allowed Tags: O, B-LOC, I-LOC, B-PER, I-PER, B-DATE, I-DATE, B-ORG, I-ORG, B-MISC, I-MISC." (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels are designed for general NLU evaluation, not administrative routing. Topic classes (sports, culture, technology, business, animals) [Q153] have no overlap with required routing categories.

OO-2: Missing categories include: cross-border worker / frontaliere issues, housing/cost-of-living, transport, tax, social security, EU institutional matters, residency permits, communal services, public health, plus a state-vs-communal routing axis and an urgency/escalation axis [regional context administrative_domain_notes; elicitation Q1, Q4].

OO-3: The voice-assistant intent labels (alarm/reminder/weather) [Q154] encode a non-administrative, non-Luxembourg-public-sector value framing — they are operational categories from a consumer assistant context, not civic communication.

OO-4: Stakeholder-driven taxonomy redesign is necessary; the elicitation confirms that routing logic requires multi-label outputs and confidence-based escalation [elicitation Q4], implying co-design with civil servants and legal experts.

OO-5: Structural validity is violated: the benchmark's single-label space cannot represent the multi-label decision structure [Q67 vs. elicitation Q4]. Content validity is violated through the absence of required administrative categories. External validity is at risk because performance on these tasks does not predict performance on the operational decision structure.

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Q153	"topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology." (p.15)

Q154	"slot_intent_detection: Detect the intent for the text given. Allowed intents: reminder/show_reminders, weather/find, reminder/set_reminder, reminder/cancel_reminder, alarm/snooze_alarm, alarm/show_alarms, alarm/set_alarm, nalarm/cancel_alarm, nalarm/time_left_on_alarm." (p.15)
WEB-12	EU AI Act Annex III Category 5(a) high-risk classification potentially relevant — implies need for documented decision rules and oversight outputs not in benchmark (https://artificialintelligenceact.eu/annex/3/)
WEB-14	Language law requires response in citizen's chosen language — implies language-of-input output dimension absent from benchmark (https://data.legilux.public.lu/filestore/eli/etat/leg/loi/1984/02/24/n1/jo/fr/html/eli-etat-leg-loi-1984-02-24-n1-jo-fr-html.html)
Q1	"This paper presents LTZGLUE, the first Natural Language Understanding (NLU) benchmark for Luxembourgish (LTZ) based on the popular GLUE benchmark for English." (p.1)
Q4	"Small and under-researched languages are particularly difficult to evaluate, as is the case with Luxembourgish (LTZ), the national language of Luxembourg, with around 400k speakers." (p.1)
Q67	"Together, the eight tasks in LTZGLUE form a broad and balanced evaluation suite, covering four binary and four multi-class settings, sentence- and document-level inputs, as well as a token-level sequence-labelling task." (p.5)
Q149	"sentiment_analysis: Classify sentiment of the text. Allowed labels: positive, neutral, negative." (p.15)

Information Gaps

- Final list of required routing categories and confidence-threshold rules

Requires Expert Verification

- Co-design of administrative topic taxonomy and multi-label/confidence scheme with civil servants, legal counsel (CNPD), and AI Act compliance officers

Remediation

Gap 1: All tasks emit single hard labels with narrow non-administrative taxonomies; no confidence scores, no multi-label outputs, no urgency scale, no language-of-input output

Recommendation: Define a deployment-specific output schema with multi-label routing categories, calibrated confidence scores tied to escalation thresholds, an urgency scale validated against civil-servant judgments, and a language-of-input field to satisfy the 1984 language law obligation [WEB-14].

ID	Evidence
WEB-14	Language law requires response in citizen's chosen language — implies language-of-input output dimension absent from benchmark (https://data.legilux.public.lu/filestore/eli/etat/leg/loi/1984/02/24/n1/jo/fr/html/eli-etat-leg-loi-1984-02-24-n1-jo-fr-html.html)